CYP Challenge direct inhibition

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Ranking: 5th (of 271 total participants)

Summary:
An ensemble of ligand-only models (Uni-Mol v2, Atom-JEPA, Chemprop and an ECFP4 network), plus physics-based scores from co-folded structures. The members are blended with non-negative weights. Single-concentration screening data is converted into pIC50 intervals for CYP1A2 and CYP2D6 so that weak compounds are represented in training.

Method

1. Ligand-only ML models as base learners
The test compounds are analogs of training compounds. Most of the signal therefore comes from ligand-only models, which effectively recall the activity of similar training compounds. To capture this signal I used the following ligand-only models:

- Uni-Mol v2 (the strongest single model)
- Atom-JEPA (a close second)
- Chemprop
- A feed-forward network on ECFP4 fingerprints

Combining several model architectures compensates for any errors or biases that each architecture might bring.

2. Co-folding + physics as a structure based orthogonal signal
ML models are known to fail on activity cliffs that the training set does not explain. To detect activity cliffs I added a structure-based signal.

For each training compound I generated 30 co-folded structures with Protenix-v2 and scored them with a physics-based scorer. This is an in-house built system that:

- First preprocesses the pose through protonation and energy minimisation.
- It then evaluates a range of physical interactions, including heme Fe binding, hydrogen bonding, van der Waals contacts, electrostatics, and π-interactions, as well as entropic contributions to arrive at a binding likelihood.

This structure signal is the final member of the ensemble. In practice the structure signal pulled down predictions where the ML models were confident but no good fit was found, and pushed up predictions where co-folding found poses that looked like strong hits. This had the most impact for CYP1A2 and CYP2D6, which both had a mix of inhibitors and inactive compounds and where the ML-only models were weaker.

3. Ensembling with non-negative weights
The members are blended per isoform with non-negative weights. The weights are fitted on the out-of-fold training predictions to minimise the soft-threshold error (the competition metric) directly, rather than the squared error. Blending compresses the spread of the predictions, so as a final step each isoform's predictions are rescaled to a wider deviation.

Data

I used the following training data sources:

- pIC50 dose response from the challenge data
- log2fc from the single concentration challenge data
- 3A4 pIC50 TDI condition from the challenge data
- pIC50 data from ChEMBL (the four isoforms + 2C19)
- Max response data from PubChem AID 1851

The ML models were trained in multi-task settings across all isoforms at once. I upweighted the competition tasks by a factor 6x compared to the other tasks.

Converting single-concentration data into dose response labels
The training and test sets differed in distribution: dose-response pIC50 was only measured for compounds that were hits in the single-concentration screen, so the training labels are enriched for strong inhibitors. The bias is largest for CYP1A2 and CYP2D6.

I addressed this for CYP1A2 and CYP2D6 by converting single-concentration screening results into pIC50 intervals:

- Compounds above a cutoff on the screen's log2 fold change were labelled as inactive. Their interval runs from the assay floor to 3.4 or 3.2 pIC50 for 1A2 and 2D6 respectively.
- The remaining screened compounds were assigned via 10 equal-count bins of log2 fold change. Each compound gets the pIC50 range (2.5th to 97.5th percentile) of the confirmed compounds in its bin.

This dataset was used throughout.

Loss function and data splitting
I used the soft-threshold MAE as training loss when fitting the ML models to align with the competition metric. I used 5-fold cross validation throughout with a butina split at ≥0.8 Tanimoto similarity. This high threshold was chosen to mimic the blind test set which contained analogs of training datapoints. As the submission set and the train set had been sampled differently it was tricky to figure out which improvements would carry over from the train set to the test set, and which would not.

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Huge thanks to the organisers and the scientists who have generously created all of this data to host this challenge. Thank you!